

Strategy vs. Confidence

Using Unobserved Data to Estimate NFL Kicker Success

Elliott Kervin | UNC Charlotte

Hypothesis

- Existing field goal models have implicit survivorship bias
 - Trained on kicks that were allowed to be attempted
- They can allow us to compare kicking results controlling for the difficulty of the kick (distance, wind, weather, etc.)
- However, there are a number of factors this does not account for:
 - The ability of the kickers that were allowed to attempt more difficult field goals and thereby their coaches confidence in them.
 - Coaching tendencies, aggressiveness, etc.
- Can we correct some of this bias by augmenting the field goal data, accounting for declined attempts?

The Data

Data Source: NFL Play-by-Play data (nflfastR) covering 10 seasons (**2015–2024**).

Primary Dataset (Attempts):

- Includes all **Field Goals** and **Extra Points**.
- Outcomes tracked: Made, Missed, and Blocked.
 - Blocks excluded from modelling analysis

Augmented Dataset (Non-Attempts):

- Captures all **4th Down Run, Pass, and Punt** plays
- *Purpose:* Establishes the "negative class" for propensity modeling.

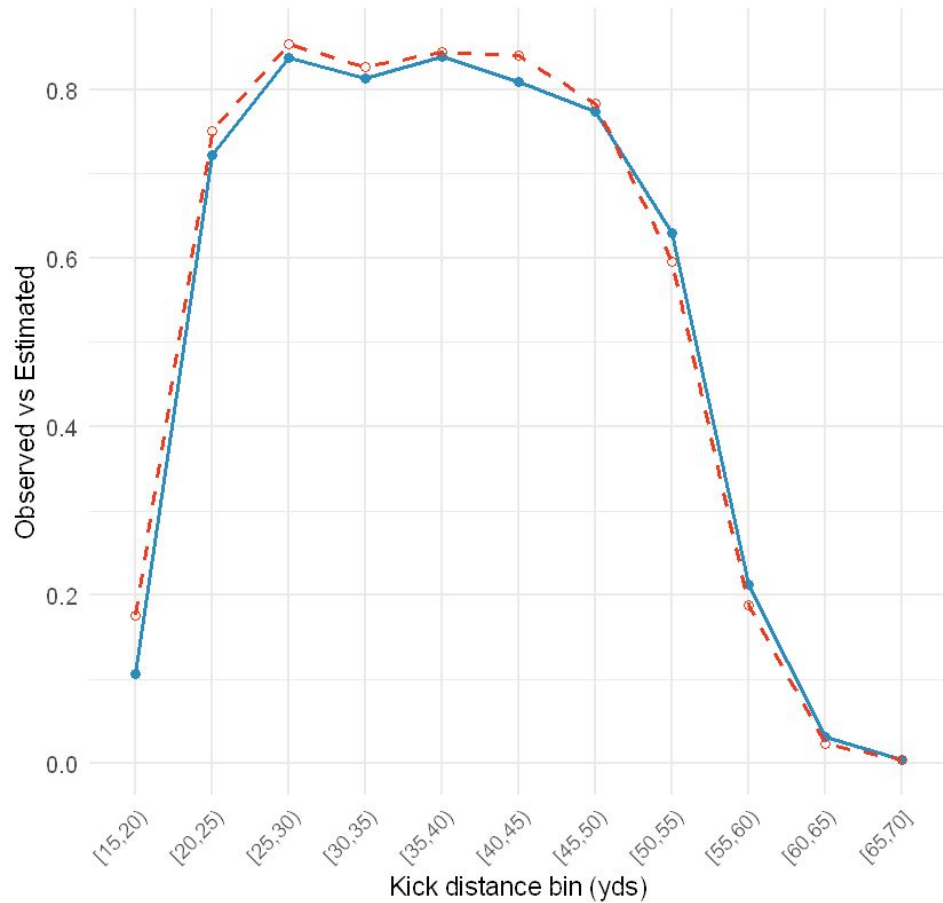
Feature Engineering:

- **Weather:** Parsed from game description text; indoor games normalized (70°F, 0 mph wind).
- **Normalization:** Environmental factors (Wind, Temp) converted to Z-scores.
- **Context:** Controls for Turf vs. Grass, Altitude (Denver, Mexico), clock running and “iced” from prior play

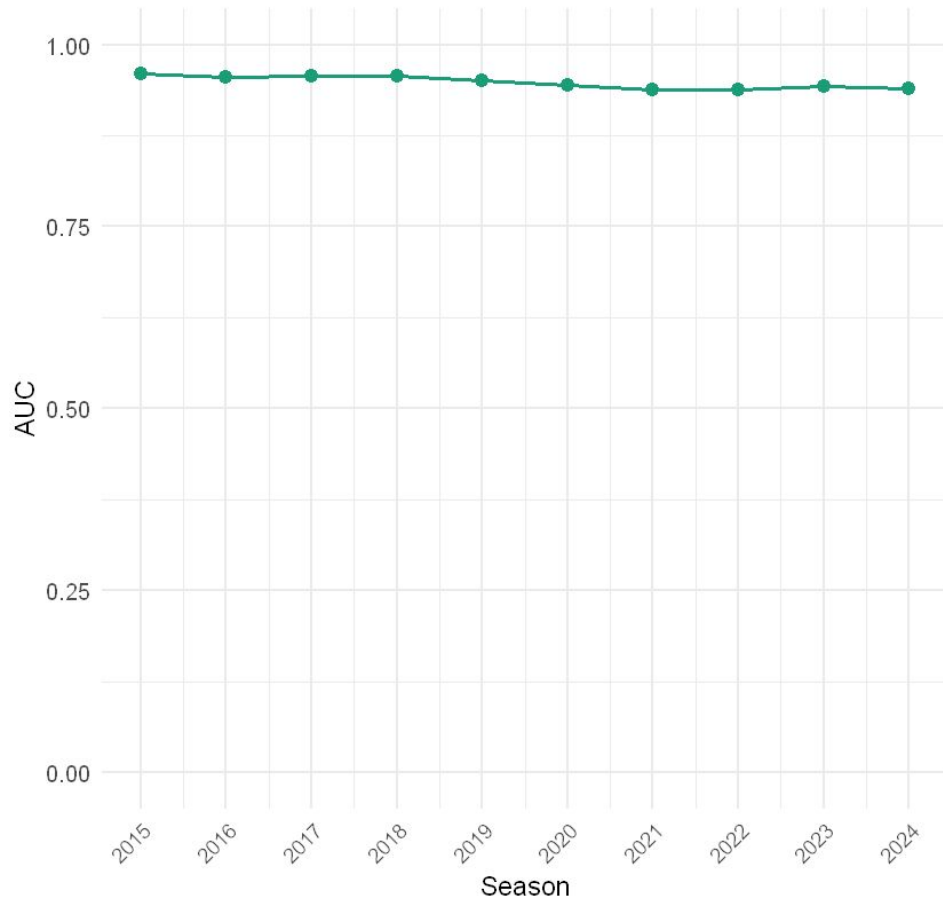
Slide: Leverage & FG Attempt Probability Modeling

- **Leverage Model (Context):**
 - Quantifies the **Win Probability (WP) at stake** for every specific kicking opportunity.
 - Two models: WP_hat_miss, WP_hat_make. Using score and time interactions.
 - $\text{Leverage} = \text{WP_hat_make} - \text{WP_hat_miss}$
 - Trained on historical play-by-play data (pre-2015) to establish a baseline for "pressure."
- **Field Goal Attempt Probability (π) Model:**
 - **Method:** Multinomial Logistic Regression predicting the probability of three outcomes: **Field Goal Attempt**, **Punt**, or **Go For It**.
 - **Training:** Rolling window approach (training on the **past 3 seasons**) to adapt to evolving coaching aggressiveness over time.
 - **Key Features:**
 - **Game State:** Leverage (WP impact), Score Differential, Time Remaining.
 - **Environment:** Wind (z-score), Temperature (z-score), Stadium effects.
 - **Geometry:** Yards to Go, Field Position.
- **Results & Performance:**
 - **High Predictive Accuracy:** The model consistently predicts coaching decisions with high precision across the decade.
 - **AUC by Season:** Range of **0.938 – 0.960**
 - **2015-2018:** ~0.960 (Highly predictable)
 - **2019-2024:** ~0.940 (Slightly lower, reflecting increased variance in "Go for It" strategies)
 - **Calibration:** Validated across distance bins (short vs. long) and game intervals to ensure robust probability estimates for the downstream IPW model.

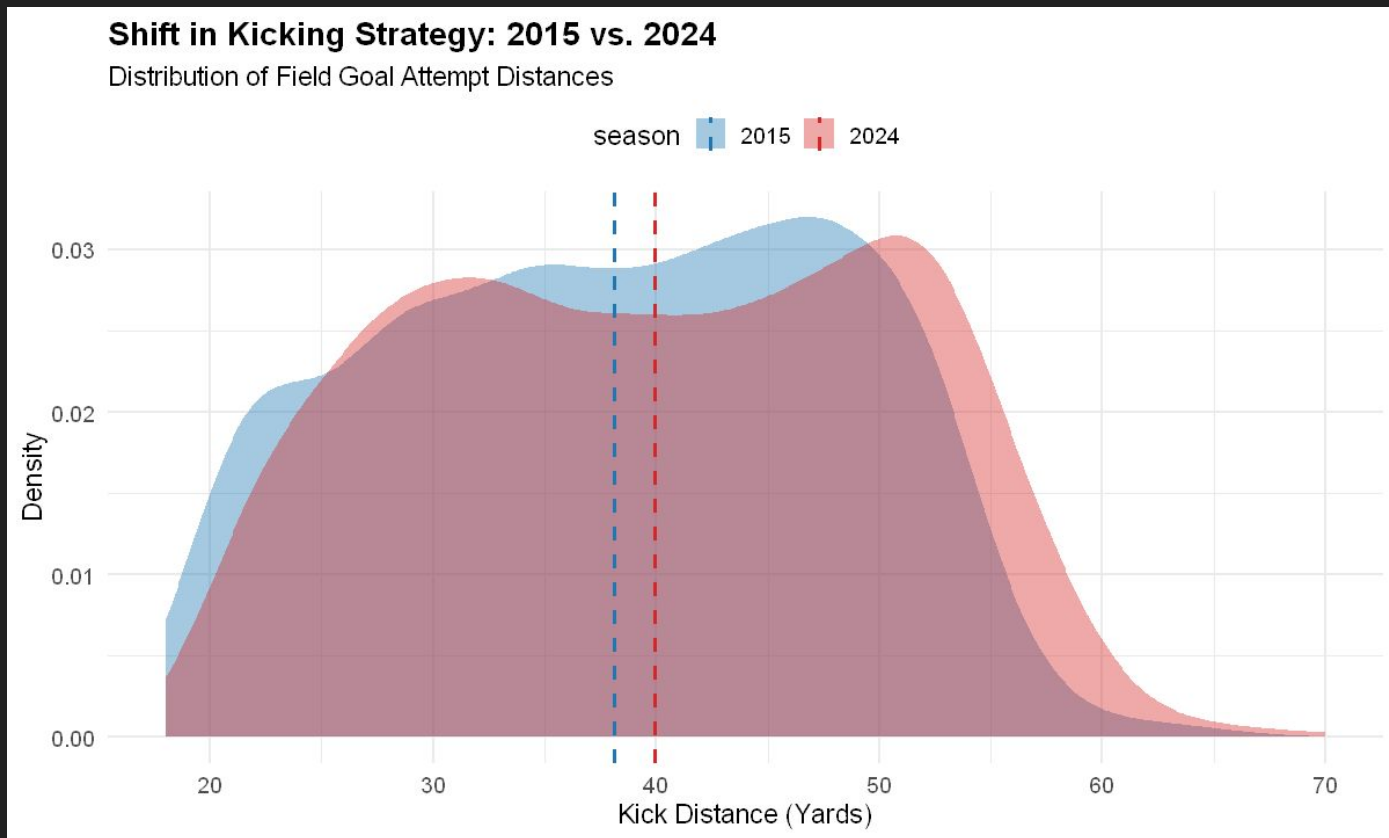
Attempt π calibration by distance (multinom)



AUC by season — Attempt π (multinom)

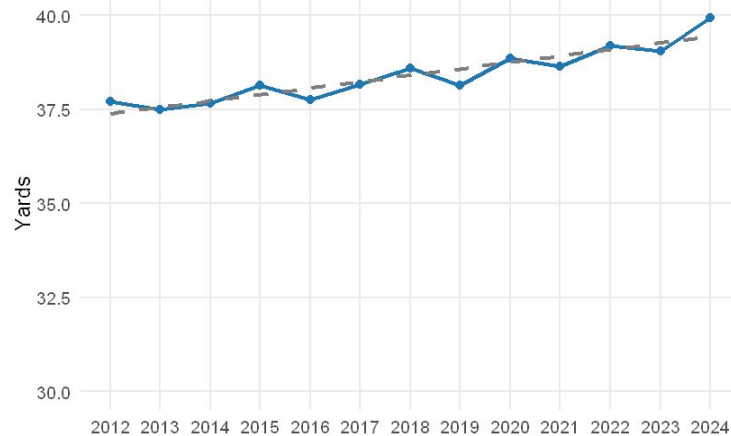


The State of the Kicker



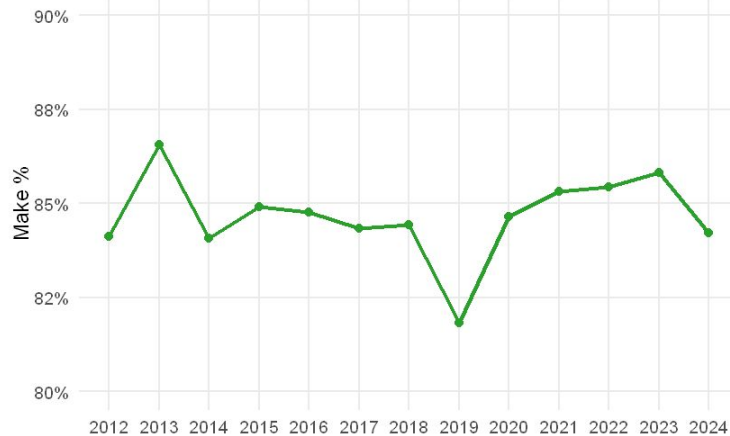
Avg Kick Distance

Mean distance of all attempts



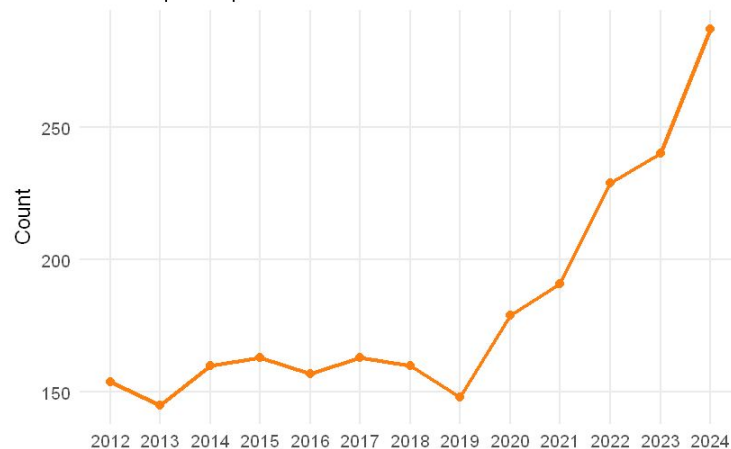
Overall Make %

Success rate for all attempts



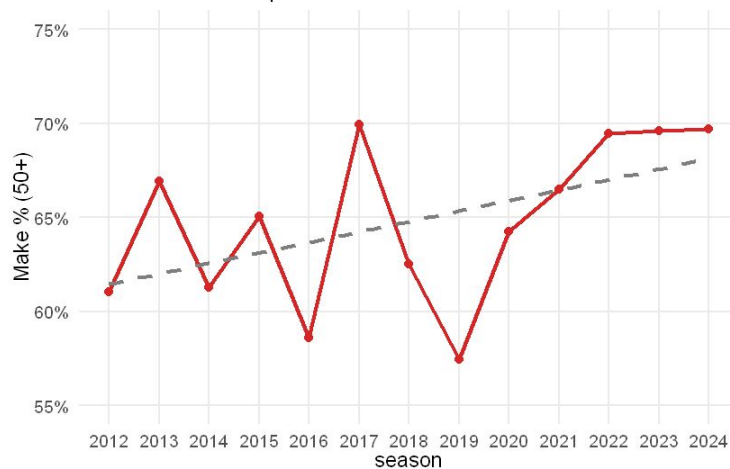
Volume: 50+ Attempts

Total deep kicks per season



Accuracy: 50+ Yards

Success rate for deep kicks



Experiment Design

M0: Distance Only (Manual Splines)

- Distance Bounds: (18, 70), Distance Knots (28, 38, 48, 58)

M1: Distance + Environmental Variables & Random Effects (Baseline)

- Standard GLMM trained on *Observed Attempts*.
- *Assumption*: "Missing data is irrelevant."

M2: Inverse Probability Weighting

- Weighted GLMM
- *Goal*: Up-weight rare attempts (long distance, high wind) to simulate the missing population.
 - Coaches could have reasonably opted to not kick in these situations, are they informative?

M3: Augmented (The "Hard Negative" Fix)

- Pseudo-Labeling: Include declined kicks (>33 yards) and treat them as **MISSES**.
- *Goal*: Penalizing kickers for non-attempts (i.e. lack of coach confidence)

Model 1 - Baseline

Structure: Generalized Linear Mixed Model (GLMM) using glmmTMB.

Target: Probability of "Making" the kick (Conditional on Attempt).

Fixed Effects:

- **Distance:** Distance Bounds: (18, 70), Distance Knots (28, 38, 48, 58)
- **Environment:** Wind (z-score), Temperature (z-score), Rain/Snow flags.
- **Game State:** Overtime, Iced, Clock Running * Timeouts
- **Interactions:** Wind * Temperature, Distance * Wind, Distance * Temp

Random Effects:

- **Stadium:** Controls for venue-specific quirks (altitude, turf).
- **Kicker-Season:** Controls for individual skill drift over time.

Formula

$$\begin{aligned}\text{logit}(P(\text{Make})) = & \beta_0 + \underbrace{\sum_{k=1}^7 \beta_k B_k(\text{Dist})}_{\text{Distance Spline}} \\ & + \beta_W \text{Wind} + \beta_T \text{Temp} + \beta_{WT} (\text{Wind} \times \text{Temp}) \\ & + \beta_{\text{Rain}} \mathbb{I}_{\text{Rain}} + \beta_{\text{Snow}} \mathbb{I}_{\text{Snow}} \\ & + \beta_{\text{OT}} \mathbb{I}_{\text{OT}} + \beta_{\text{Ice}} \mathbb{I}_{\text{Iced}} \\ & + \beta_C \text{Clock} + \beta_{TO} \text{Timeouts} + \beta_{CT} (\text{Clock} \times \text{TO}) \\ & + \text{Wind} \cdot \sum_{k=1}^7 \delta_k B_k(\text{Dist}) \quad (\text{Dist} \times \text{Wind}) \\ & + \text{Temp} \cdot \sum_{k=1}^7 \gamma_k B_k(\text{Dist}) \quad (\text{Dist} \times \text{Temp}) \\ & + u_{\text{Stadium}} + v_{\text{Season:Kicker}}\end{aligned}$$

Model 2 - Inverse Probability Weights

Objective: Create robust Inverse Probability Weights that correct for selection bias while minimizing excessive variance or numerical instability.

Step 1: Clipping & Stabilization

- **Boundaries:** Predicted FG probabilities (π) are clipped to [0.02, 0.98]
- **Stabilized Weights:** mean FG attempt propensity / (π).
 - Within each season
 - Inverse for declined attempts $(1 - \text{FG}) / (1 - \pi)$
- Weights are then scaled within each season so the mean weight is exactly **1.0**.

Step 2: Outlier Protection (The "3-Sigma" Cap)

- **Capping:** Weights exceeding **Mean + 3 SD** are truncated.
- **Re-Normalization:** Weights are re-scaled post-capping to preserve the total Effective Sample Size (ESS).

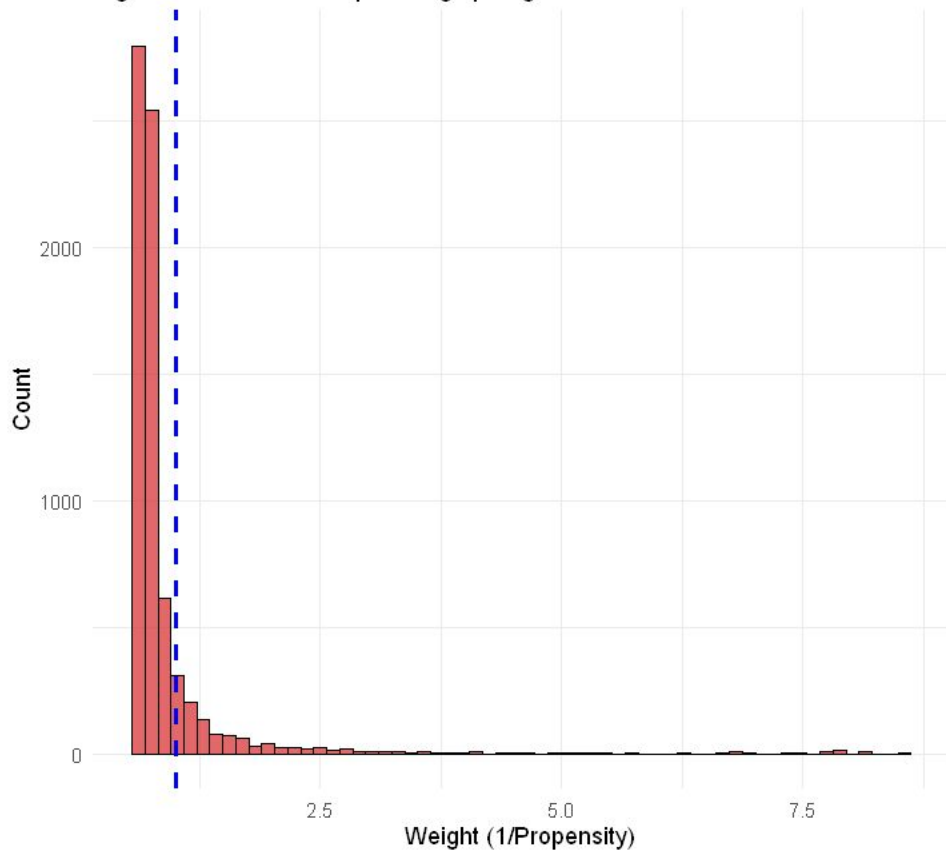
```
--- IPW Weight Statistics ---
```

```
# A tibble: 1 × 8
```

	Total_N	Min_W	Median_W	Mean_W	Max_W	SD_W	ESS	ESS_Ratio
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	10432	0.617	0.698	0.944	8.55	0.875	5612.	0.538

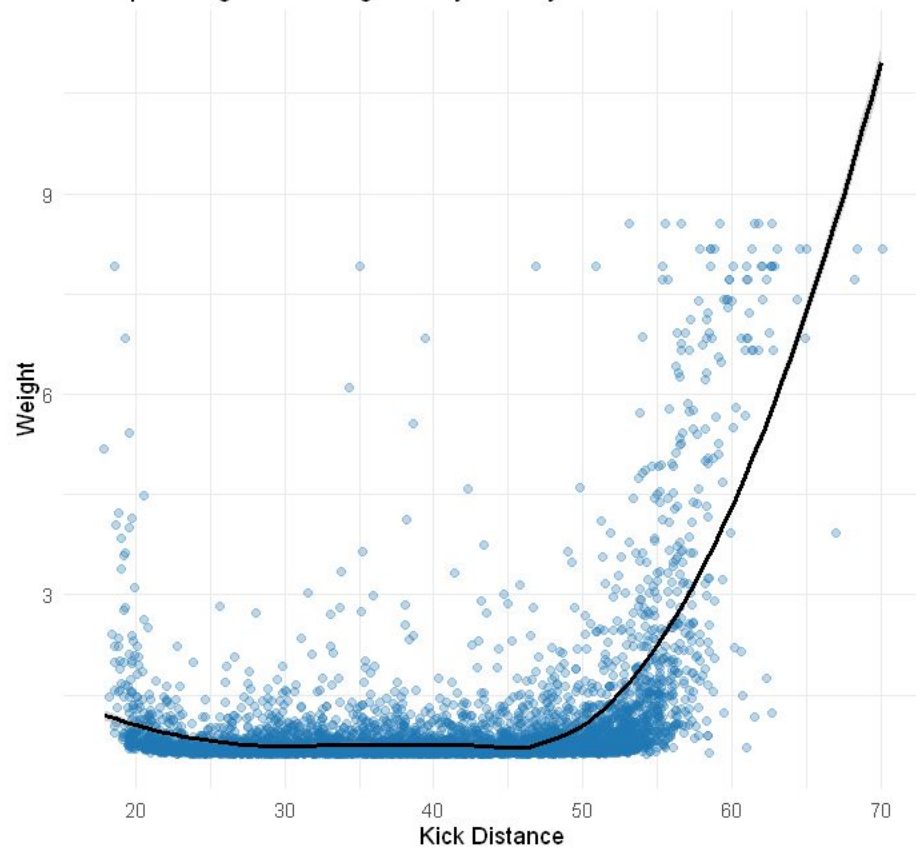
Distribution of Weights

Right skew = Rare attempts being upweighted



Weights vs. Distance

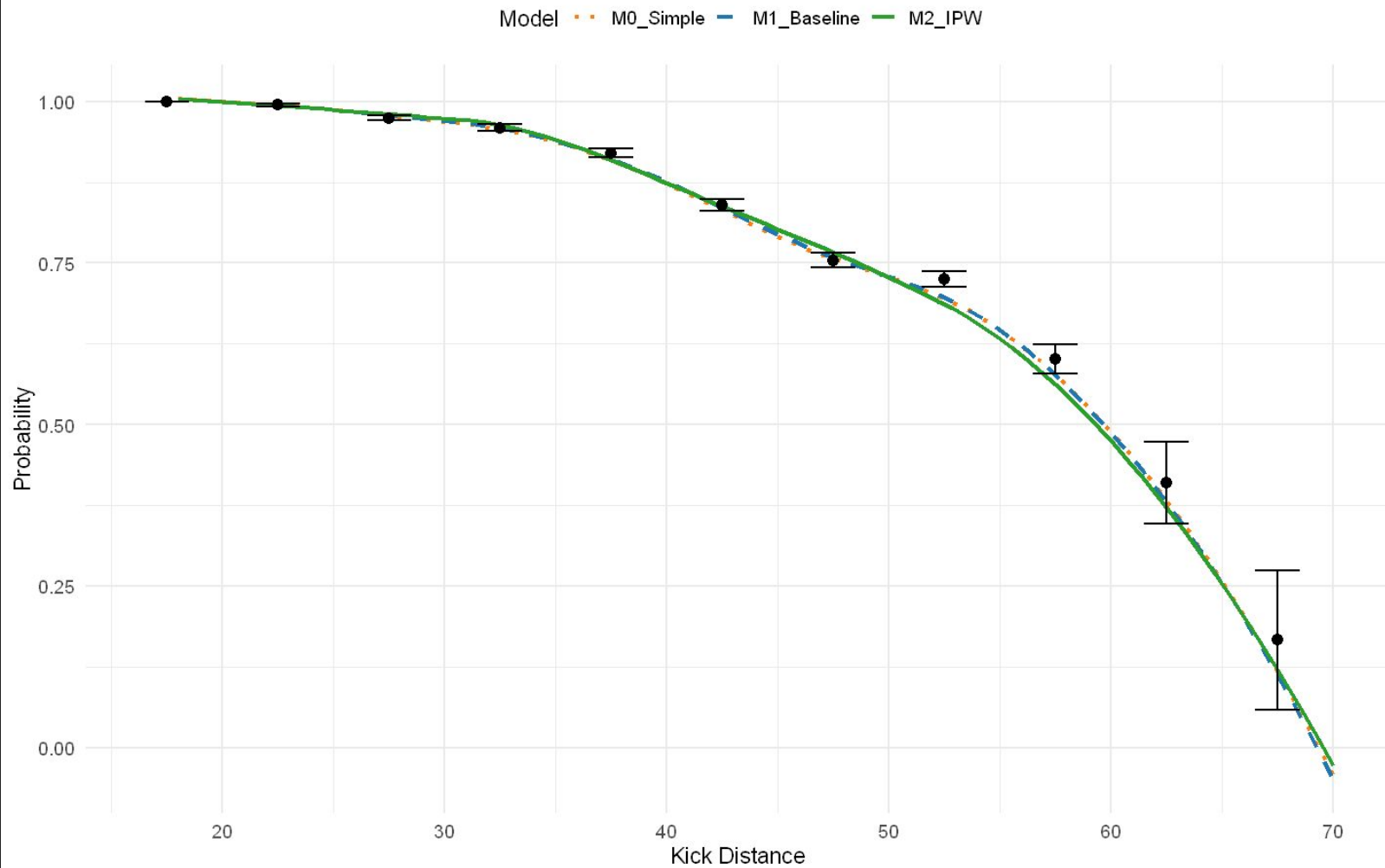
We expect weights to rise significantly at 50+ yards



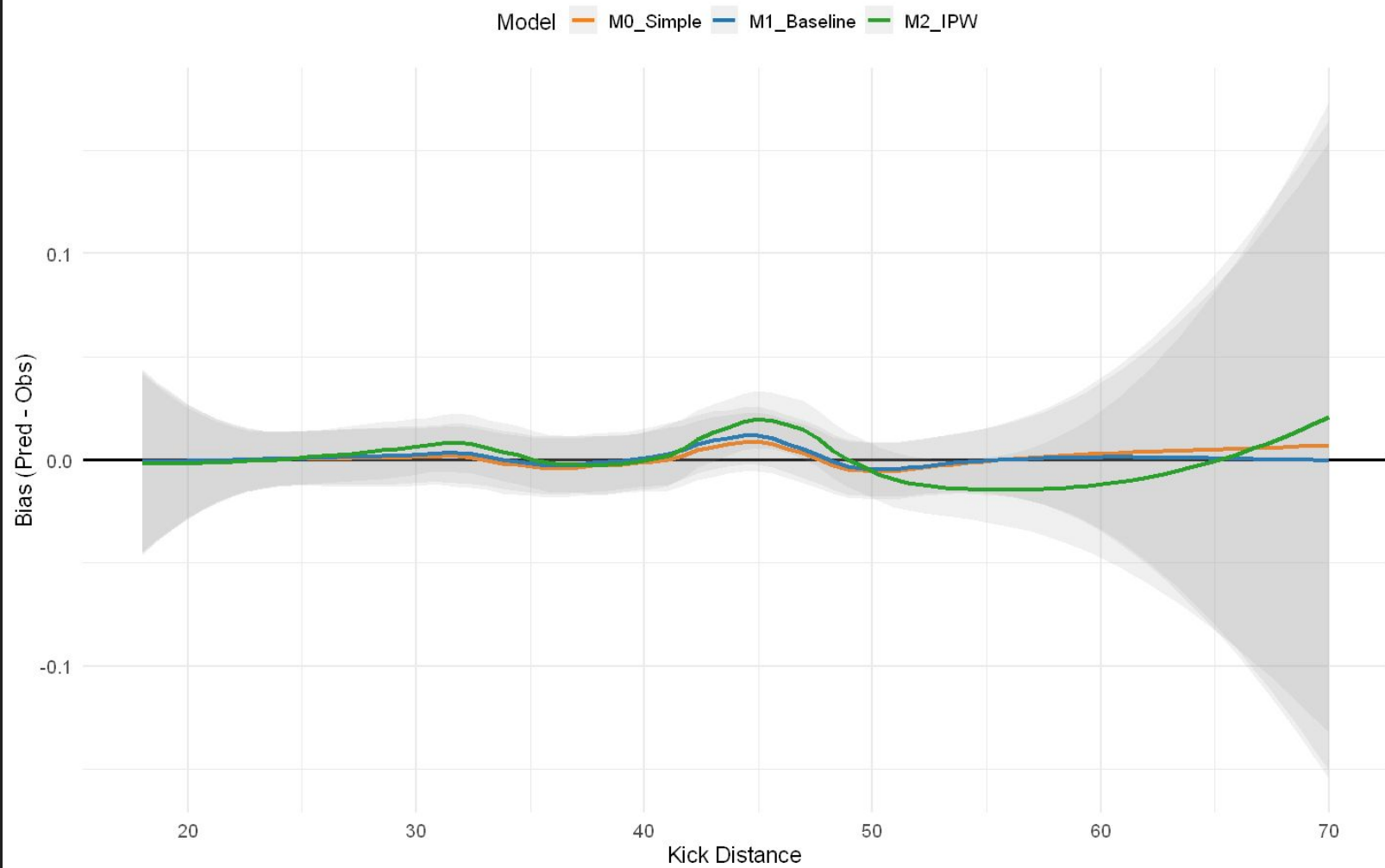
Metric Comparison (In Sample)

Model	brier	logloss	auc	calib_error	accuracy	precision	recall
M0 (Distance Only Full)	0.105	0.34	0.775	0.007	0.863	0.865	0.997
M1 (Baseline)	0.101	0.325	0.808	0.016	0.864	0.866	0.996
M2 (IPW)	0.099	0.321	0.808	0.012	0.871	0.874	0.994

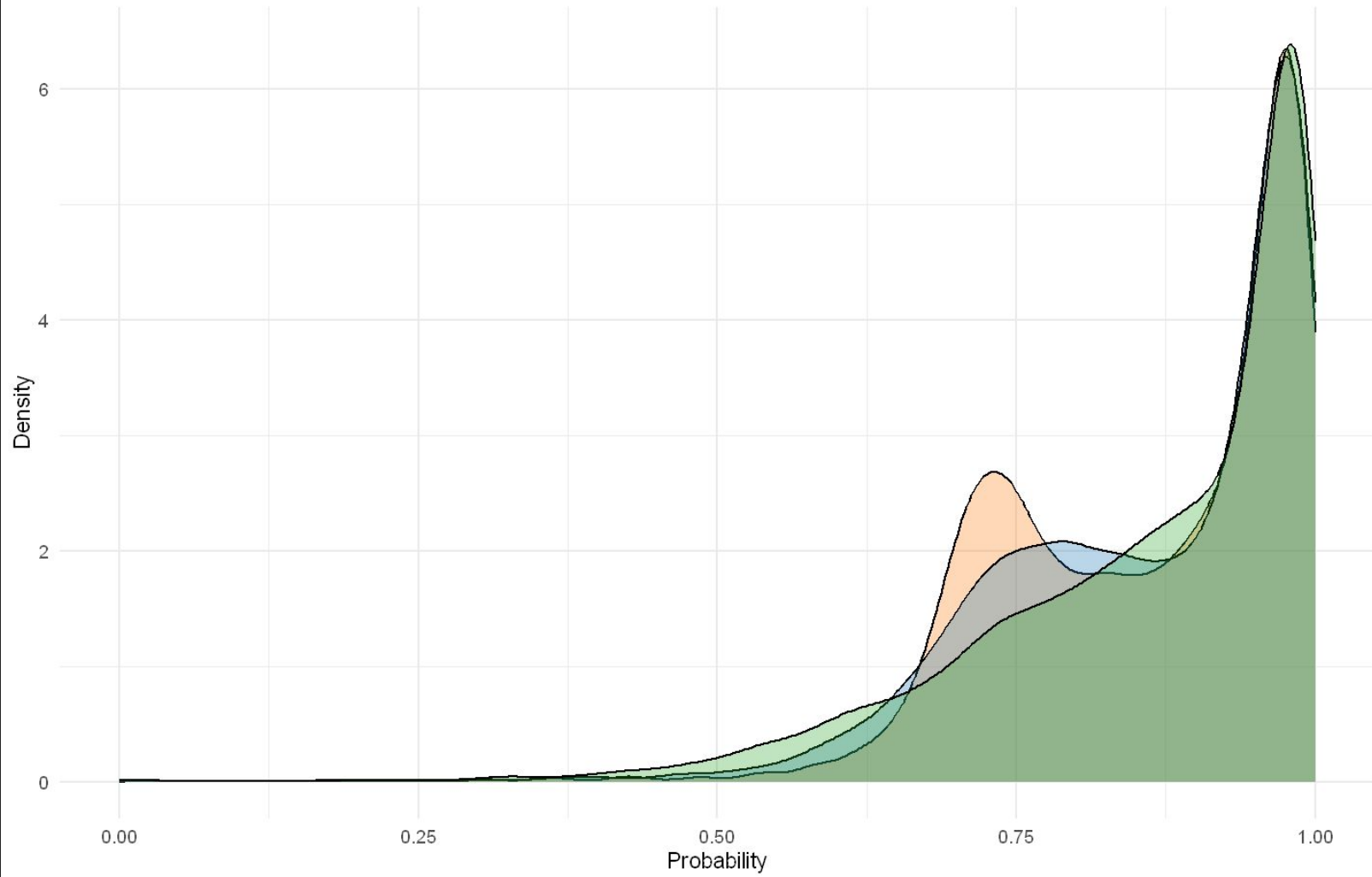
Truth Curve: Predicted vs Actual



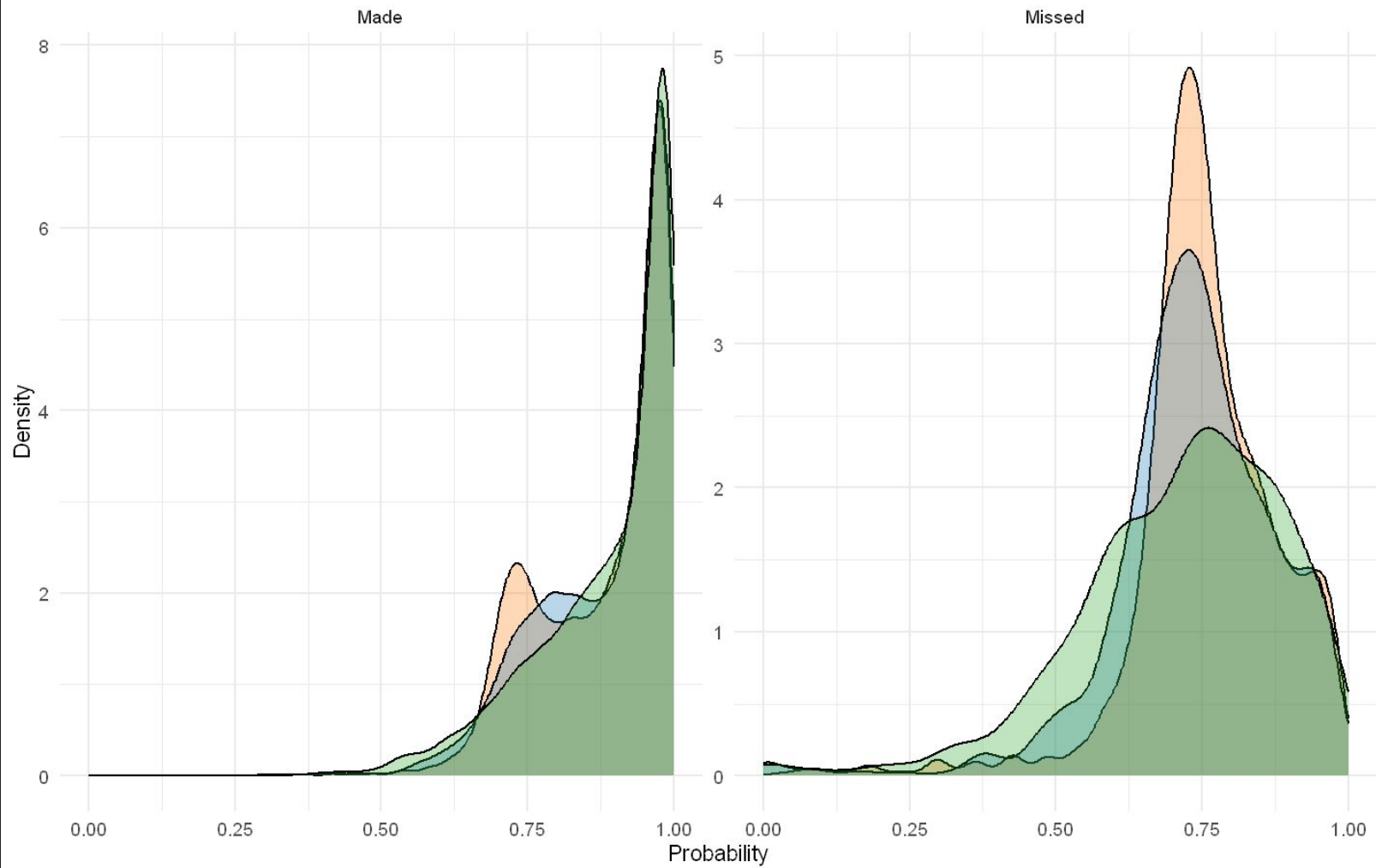
Bias Curve: Residual Analysis



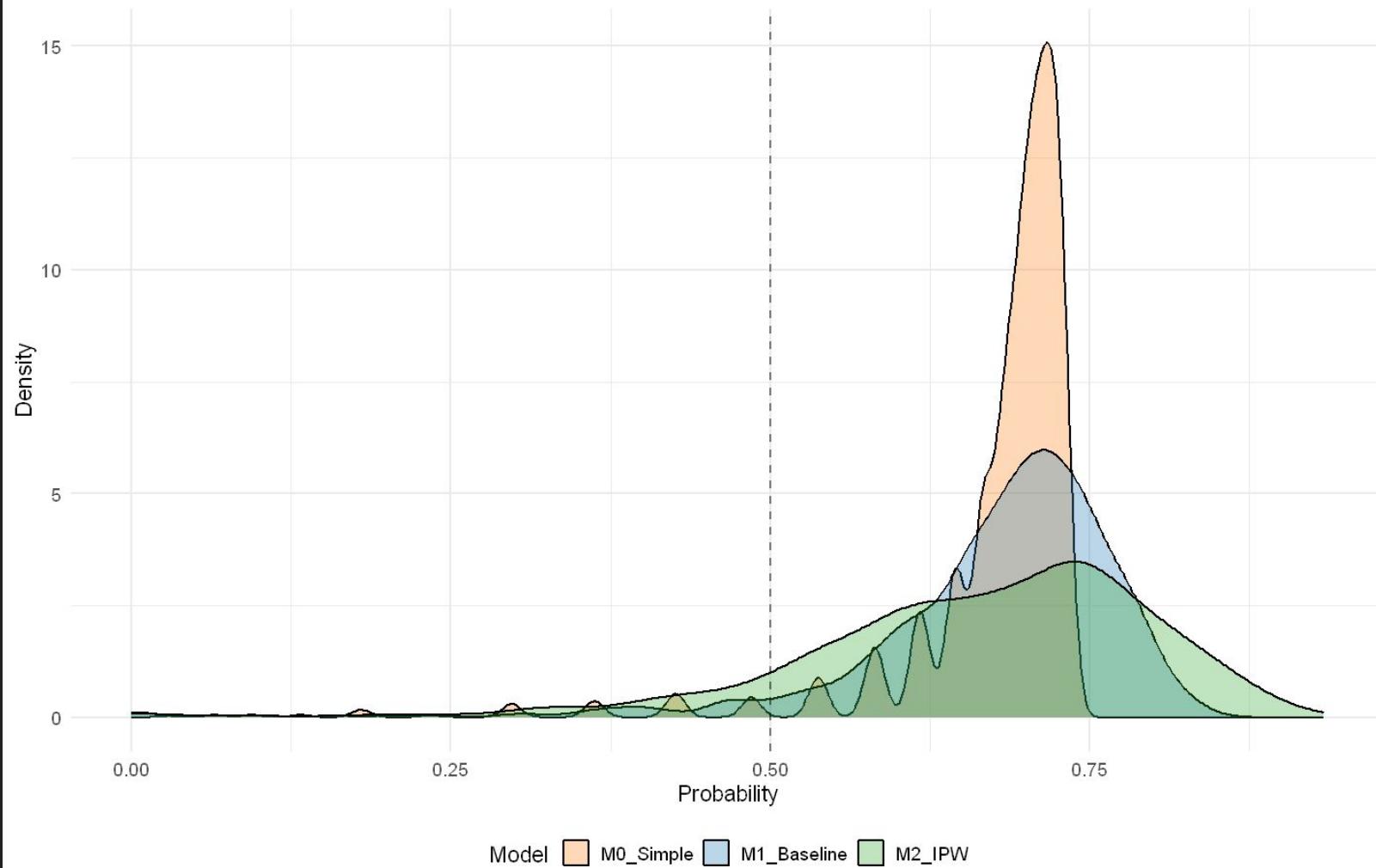
Global Probability Density



Separation Analysis



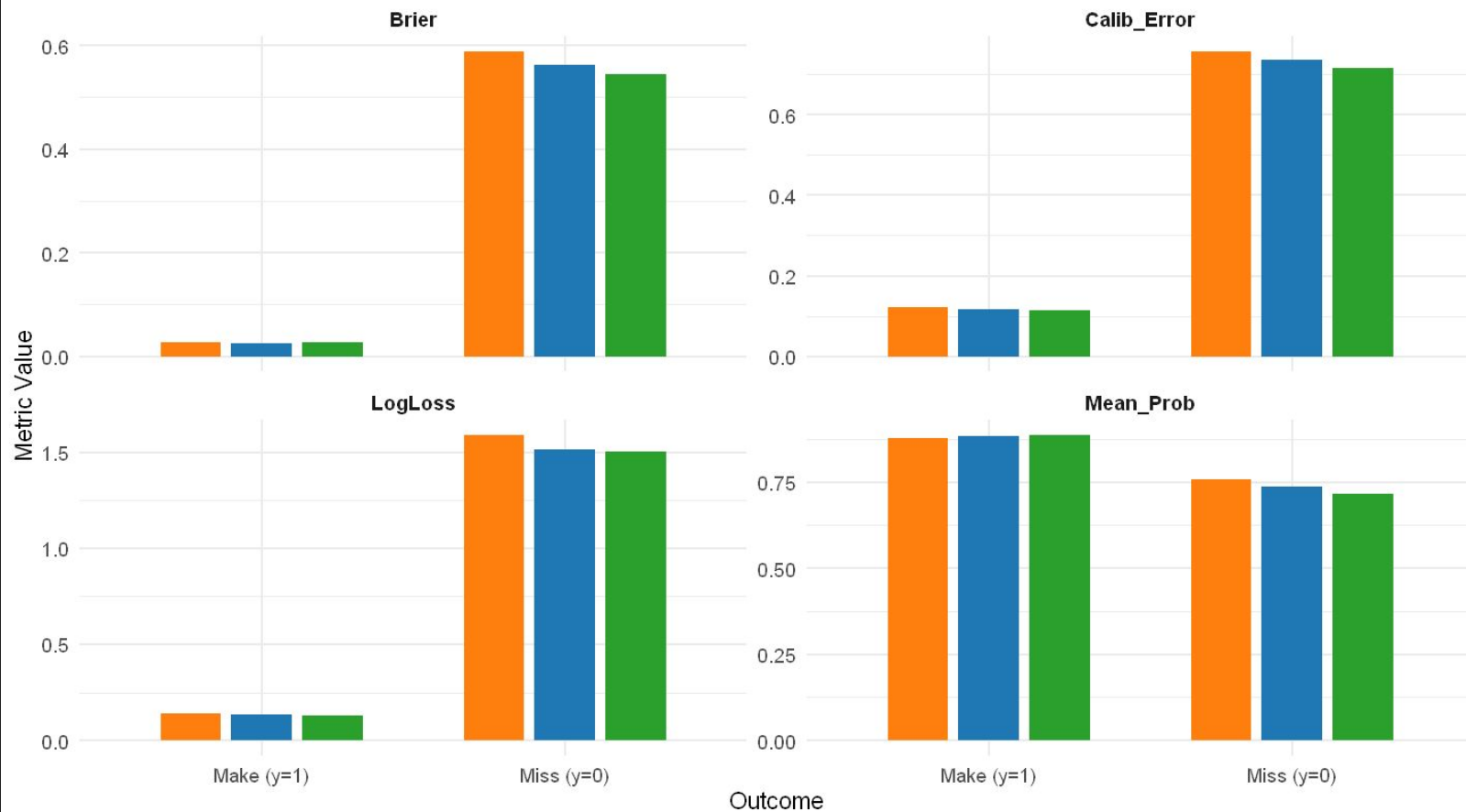
Tail Shift (50+ Yards)



Model Weakness Analysis: Makes vs. Misses

Comparing performance on successful vs. failed kicks separately

Model M0_Simple M1_Baseline M2_IPW



Metric Comparison (Out of Sample)

Subset	brier	logloss	auc	calib_error	accuracy	precision	recall
M1 (Global)	0.098	0.322	0.772	0.018	0.876	0.879	0.995
M2 (Global)	0.1	0.331	0.76	0.027	0.872	0.882	0.986
M1 (50+ yds)	0.188	0.561	0.653	0.08	0.737	0.746	0.97
M2 (50+ yds)	0.193	0.572	0.641	0.076	0.717	0.752	0.912
M1 (Prob. 25-75%)	0.201	0.591	0.568	0.05	0.715	0.726	0.972
M2 (Prob. 25-75%)	0.206	0.603	0.571	0.073	0.697	0.731	0.917

Model 3 - PAT & Declined Attempt Augmentation

Objective: Address sample scarcity and survivorship bias by expanding the training set to include **PATs** (to anchor short-range probability) and **Declined Attempts** (to capture latent difficulty).

Experimental Design (Pseudo-Labeling):

- **Inclusion Logic:** Iteratively tested filters for Attempt Propensity (π) and Distance to isolate the "Decision Zone" (excluding obvious punts/short "go-for-it" situations).
- **Outcome Imputation:** Experimented with **Hard Negatives** (Assume Miss) vs. **Threshold Positives** (Assume Make if $P(\text{Make}) > T$) to determine the label for non-attempts.
- **Weighting Schemes:** Varied the influence of augmented data from aggressive ($w=1.0$) to conservative ($w=0.05$).

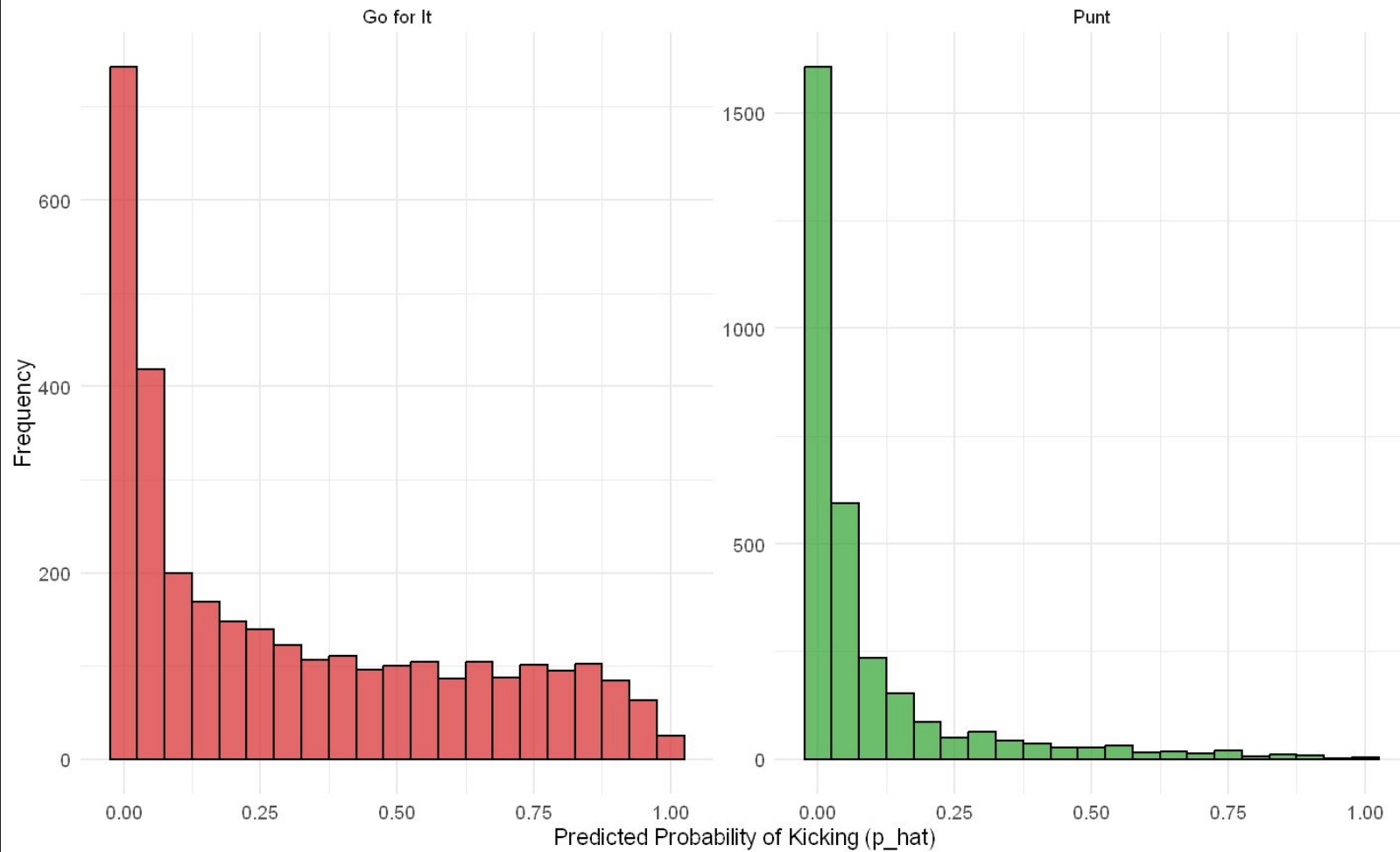
The "Champion" Configuration:

- **Subset:** Non-attempts with $\pi \geq 25\%$ and **Distance > 33 yards**. All declined attempts pseudo-labelled as miss.
- **Weighting:** Included alongside PATs at a **small fixed weight (0.10)** to provide signal without overwhelming observed data.

Key Findings:

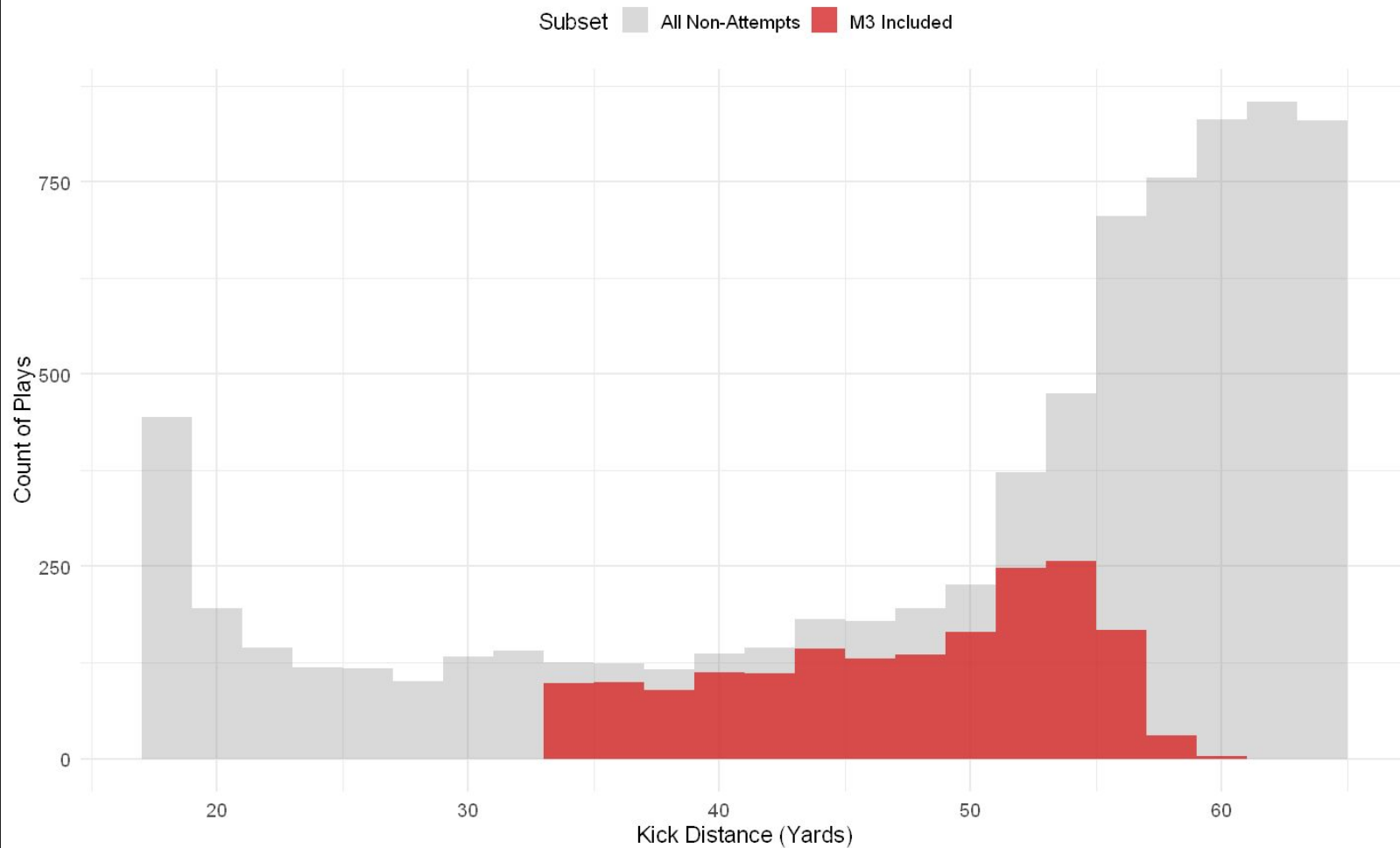
- **Global Performance:** No augmented combination meaningfully outperformed the Baseline (M1) on Brier Score or AUC.
- **Weight Sensitivity:** **Lower weights** consistently performed better than high weights. Heavy weighting of pseudo-labels degraded calibration, confirming that augmented data introduces more noise than signal compared to the "Rational" observed dataset.

Distribution of Kick Propensity (p_{hat}) by Decision Type
Comparing 'Punts' vs 'Go for It' decisions in the Non-Attempt dataset

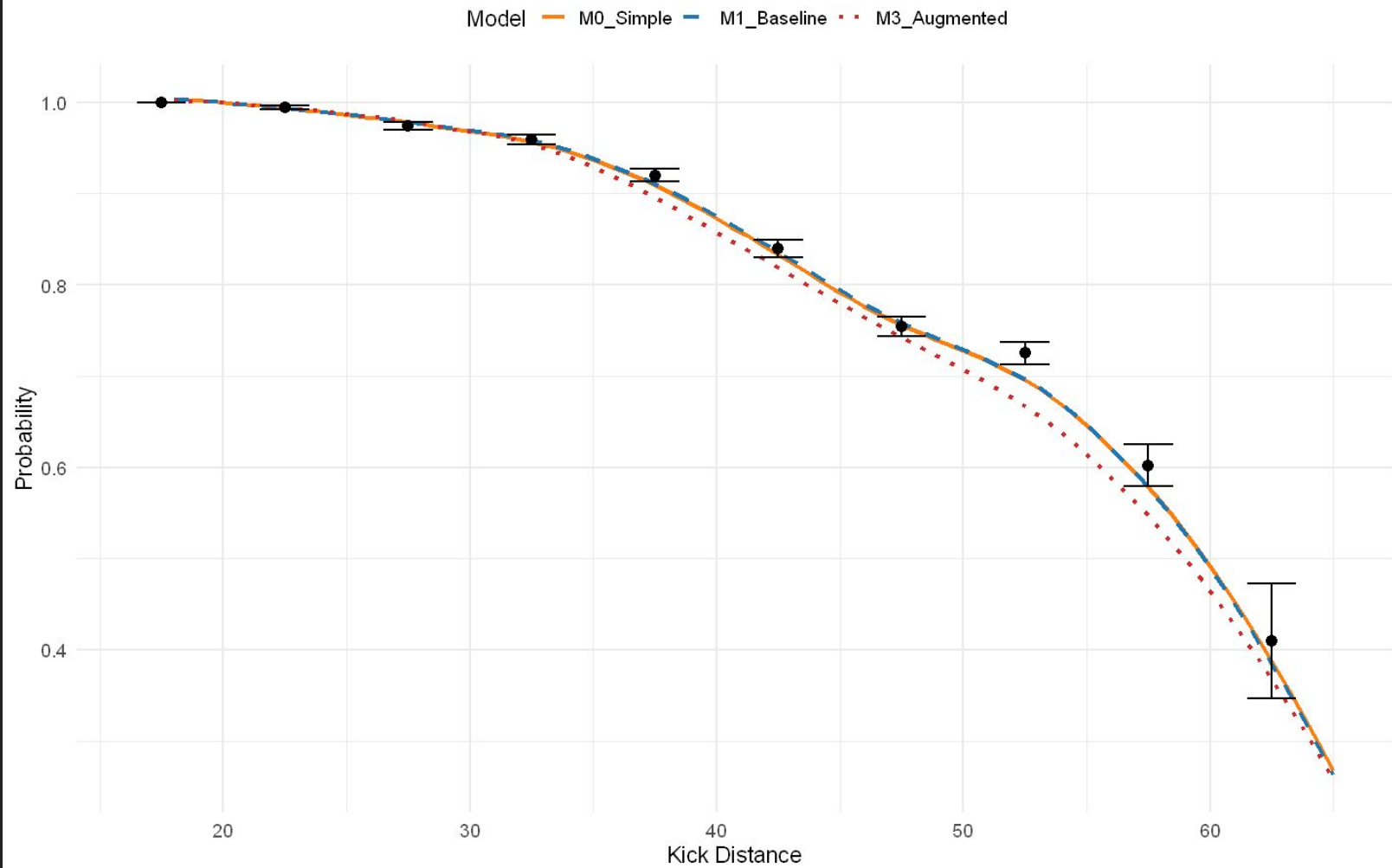


Volume Analysis: Which distances survive the filter?

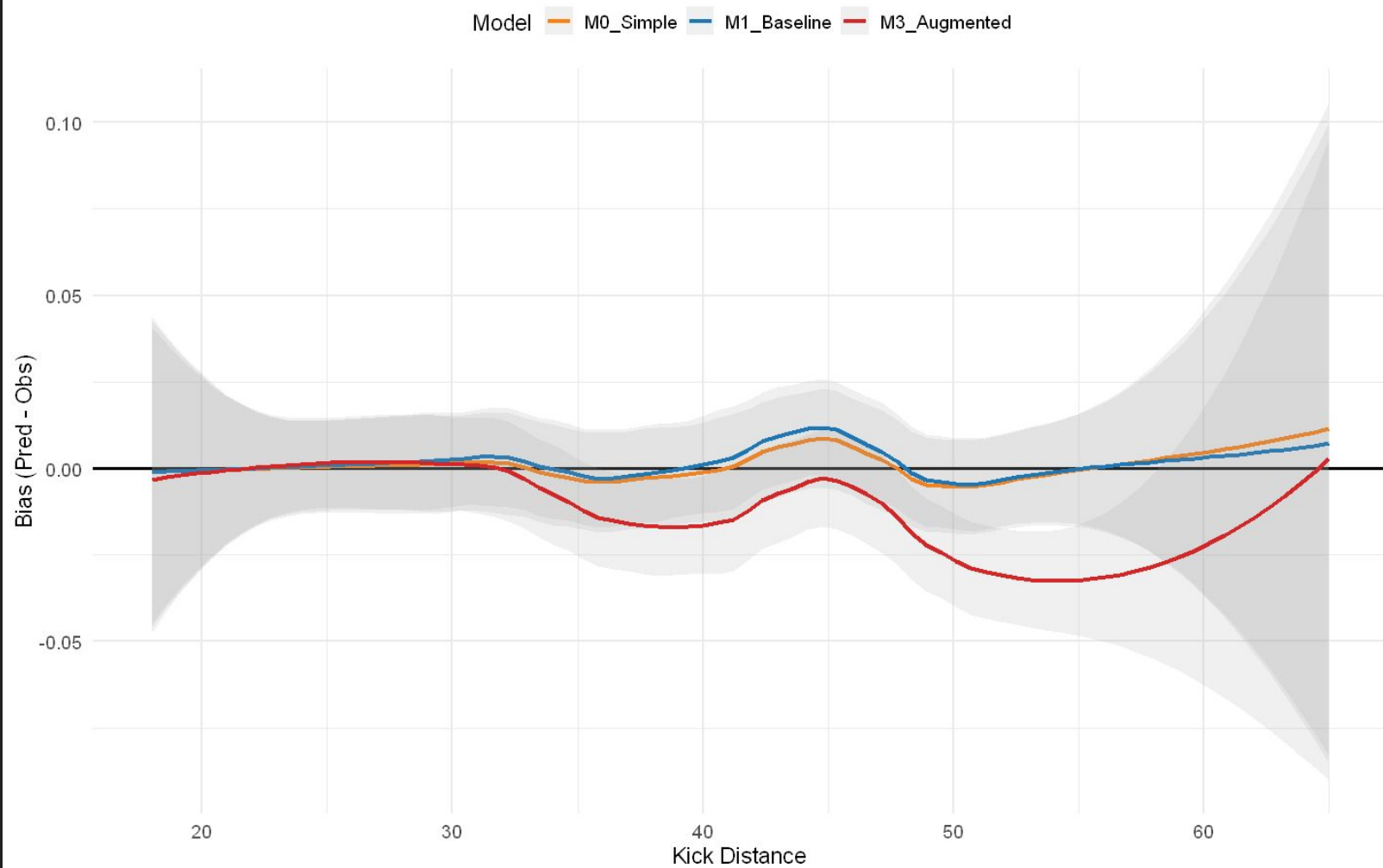
Note the exclusion of short 'Go-for-it' plays (<33y) and low attempt probability kicks (<25%)



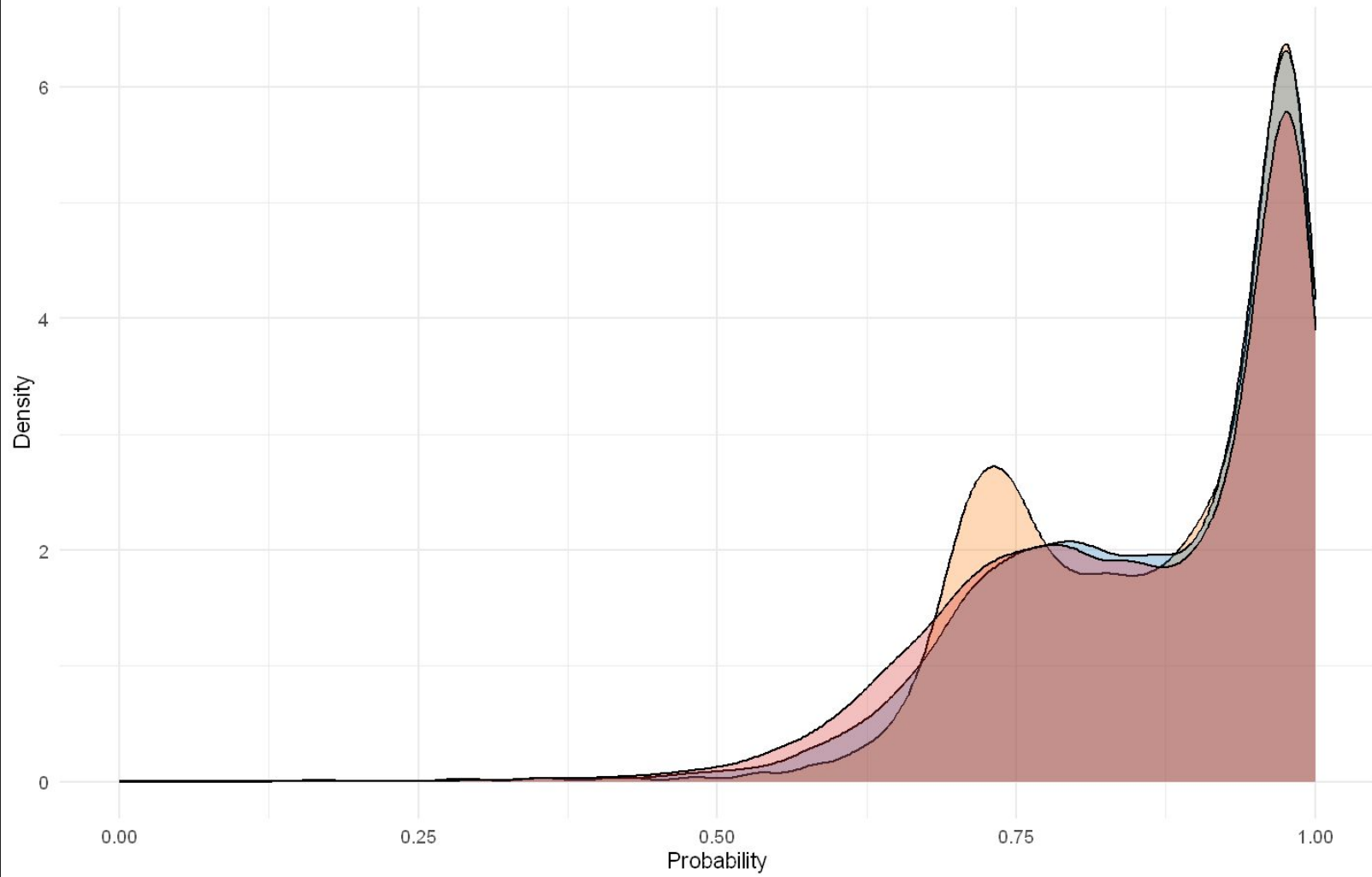
Truth Curve: Augmented vs Baseline



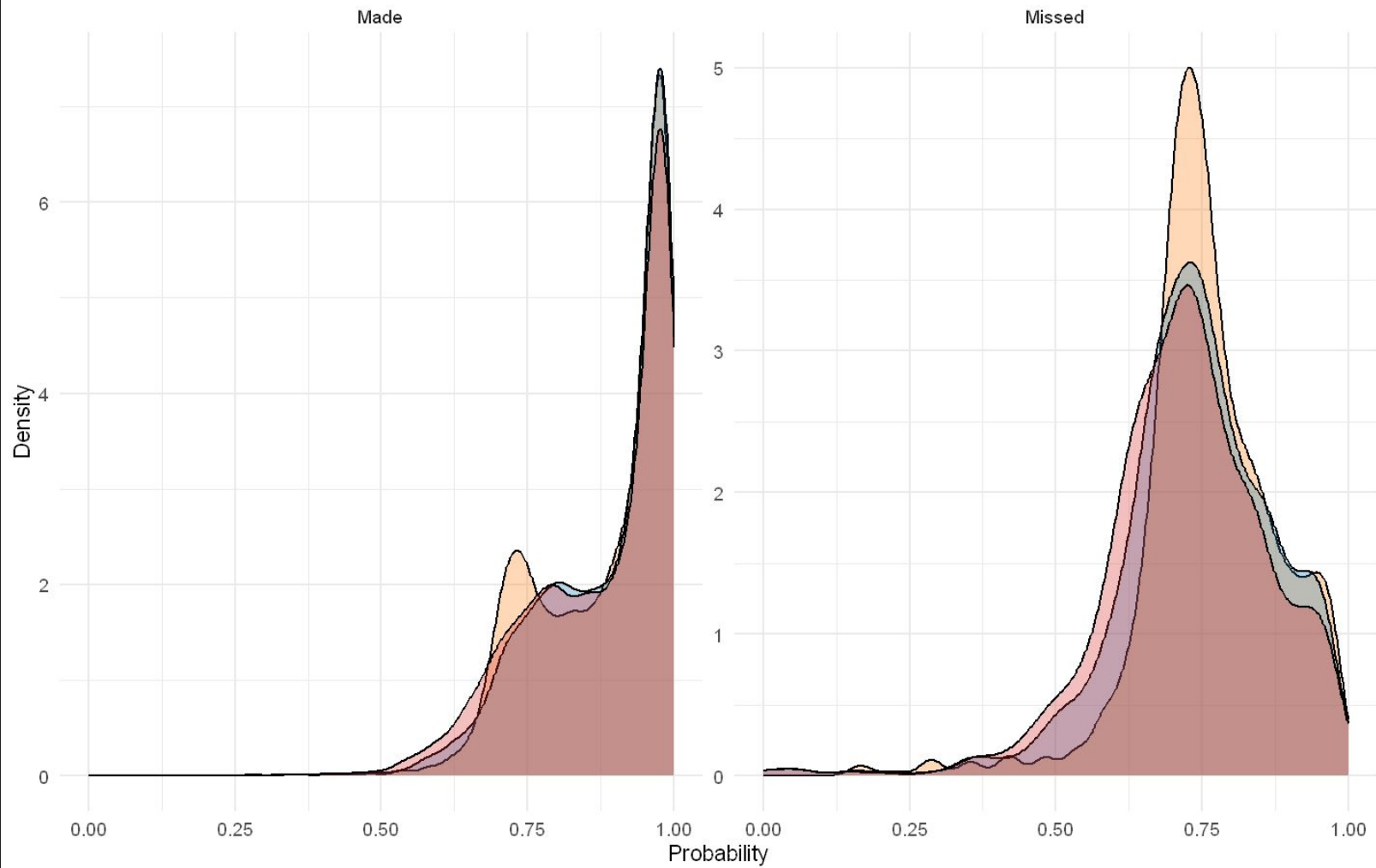
Bias Curve: Residual Analysis



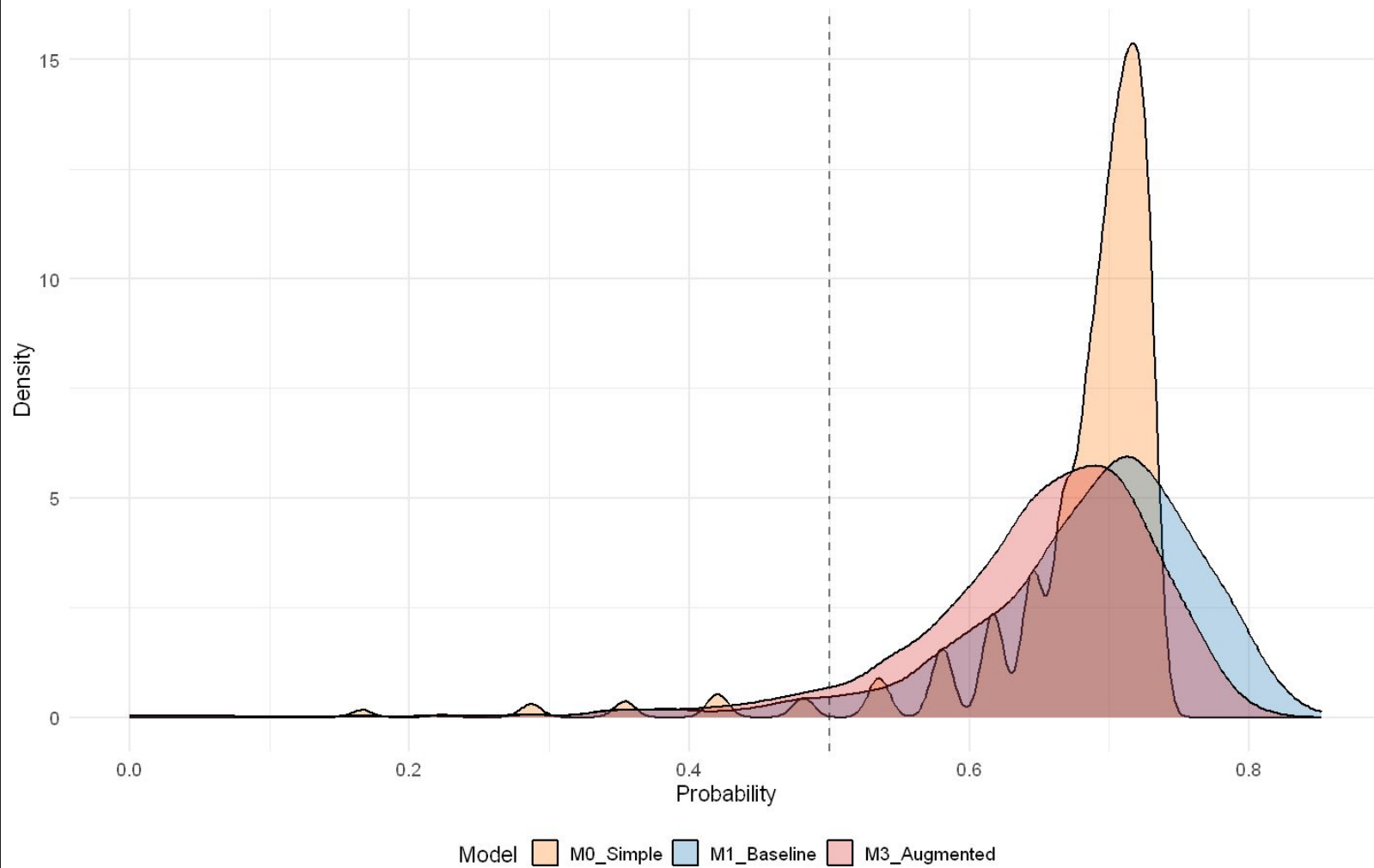
Global Probability Density



Separation Analysis



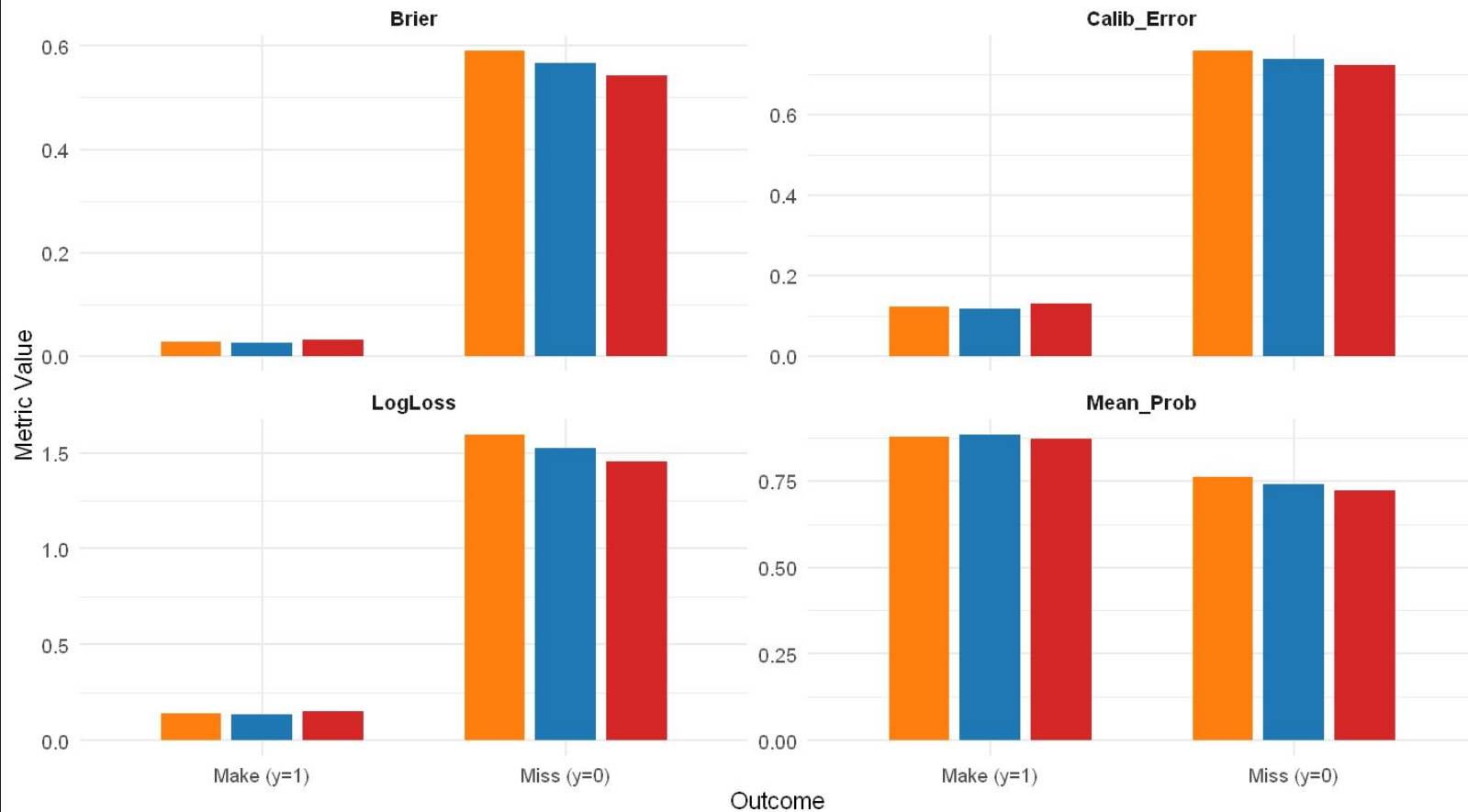
Tail Shift (50+ Yards)



Make vs. Miss Analysis: Augmented Model Impact

Does the Augmented model (Red) reduce error on Misses?

Model M0_Simple M1_Baseline M3_Augmented



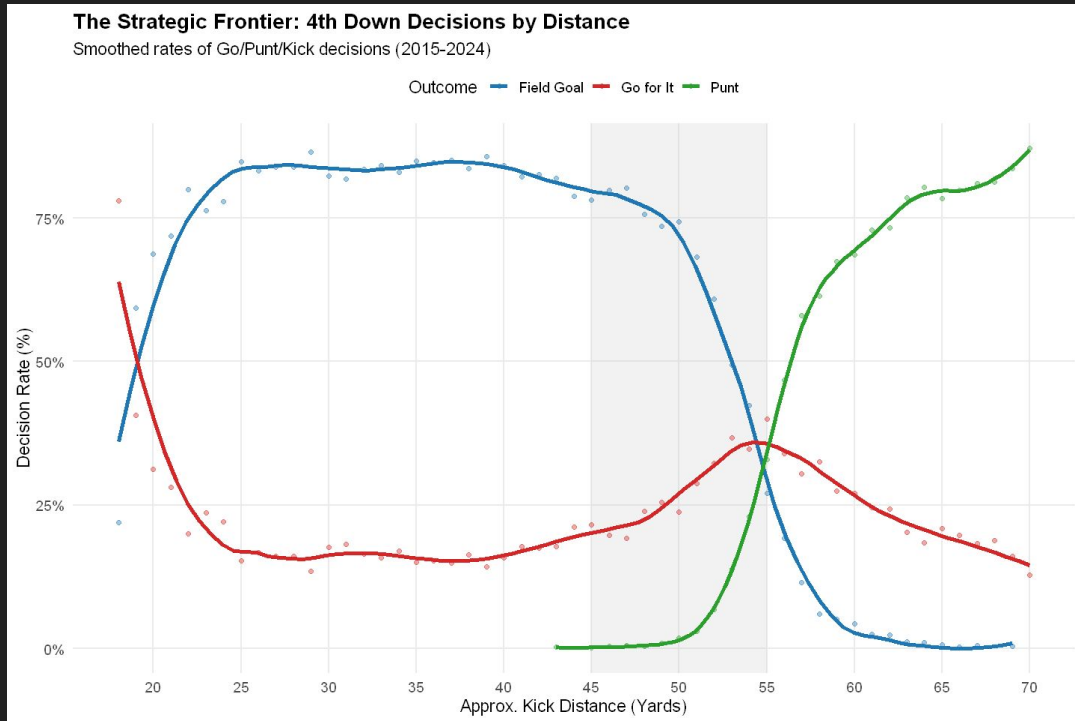
Takeaways

- **IPW Performance:** Correcting for selection bias (IPW) provided **no significant global performance increase** (Brier[M2] $\approx$ Brier_M1). The "biased" reality (M1) is the most predictable one.
- **The 'Tail' Signal:** However, IPW *did* improve discrimination in the deep tail (50+ yards) by becoming more **pessimistic**. It learned that long kicks are harder than the raw (filtered) data suggests. Overall calibration still worse than M1.
- **The 'Unicorn' Effect:** This pessimism was NOT largely driven by "desperation" (end-of-game heaves), as the propensity model already accounts for these. It was driven by **"Unicorn" attempts**—rare, long-distance kicks taken in standard game states (e.g., Q1/Q2) where coaches typically punt or go for it.

Season	Team	Dist	Score Diff	Qtr	Time Rem.	Weight	Made
2017	NO	56	+14	4	9:02	8.55	0
2017	BAL	62	-19	2	30:02	8.55	0
2017	ATL	59	+17	2	30:02	8.55	0
2017	DEN	53	-14	2	41:10	8.55	1
2017	NE	62	+14	2	30:05	8.55	1
2017	GB	57	+7	3	19:20	8.55	0
2017	LA	63	+7	2	30:02	8.55	0
2024	DAL	65	-7	1	51:38	8.17	1
2024	MIN	58	+24	4	4:48	8.17	1
2024	KC	65	-3	2	30:04	8.17	0

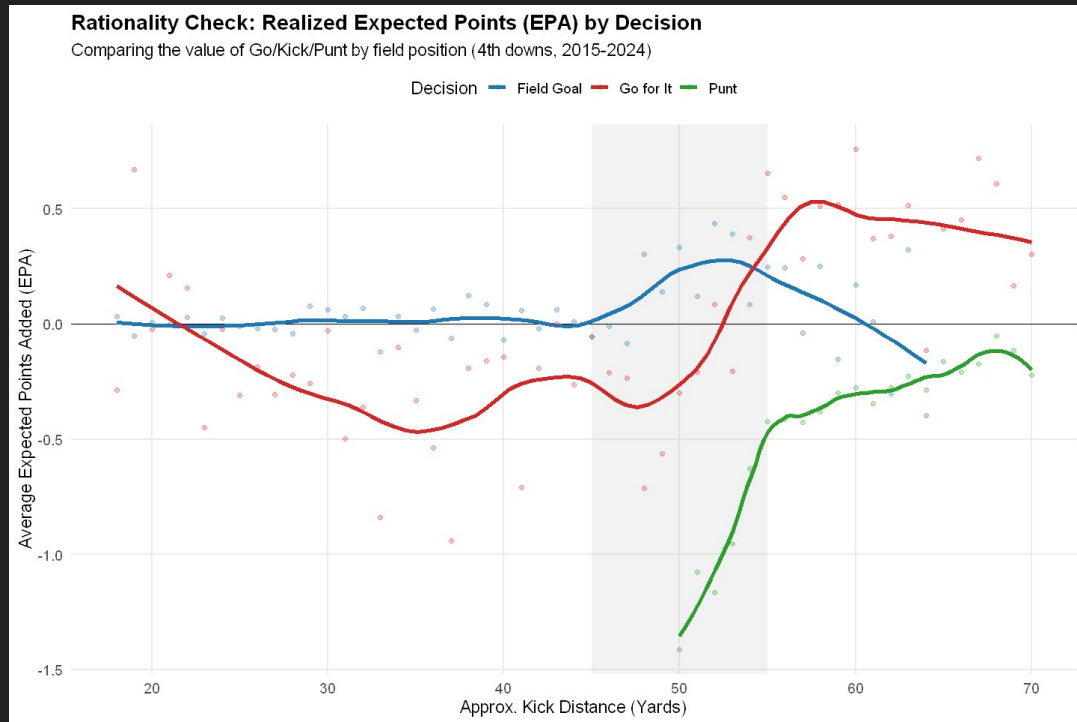
Takeaways

- **The Decision Zone:** The 45–55 yard range is "No Man's Land." This is where all play options converge
- Punting is inefficient (net ~20 yards), and Kicking is declining in value (3 points * declining make rate).
- **Rationality Confirmed:** Coaches decline kicks in this zone **not** because they lack confidence in the kicker's leg strength, but because **maximizing Win Probability requires aggression**.
- **Modeling Implication:** Treating these non-attempts as "Misses" (Pseudo-Labeling) is structurally incorrect. It penalizes the kicker for the coach's correct strategic choice to chase a Touchdown.



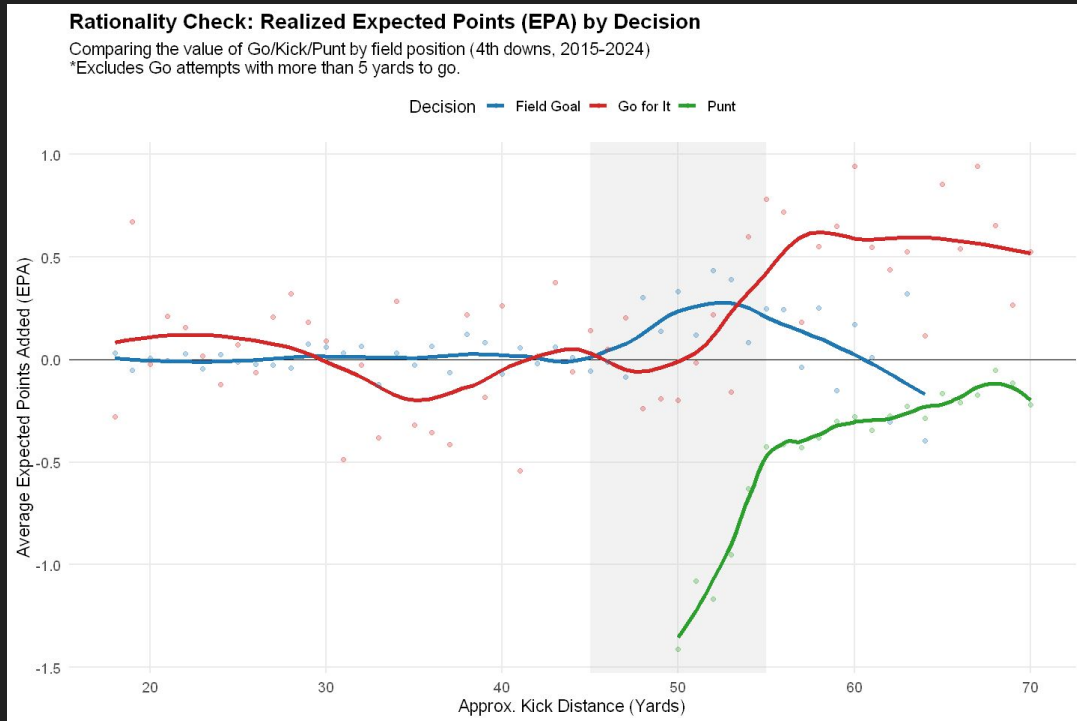
Takeaways

- **The Decision Zone:** The 45–55 yard range is "No Man's Land." This is where all play options converge
- Punting is inefficient (net ~20 yards), and Kicking is declining in value (3 points * declining make rate).
- **Rationality Confirmed:** Coaches decline kicks in this zone **not** because they lack confidence in the kicker's leg strength, but because **maximizing Win Probability requires aggression**.
- **Modeling Implication:** Treating these non-attempts as "Misses" (Pseudo-Labeling) is structurally incorrect. It penalizes the kicker for the coach's correct strategic choice to chase a Touchdown.



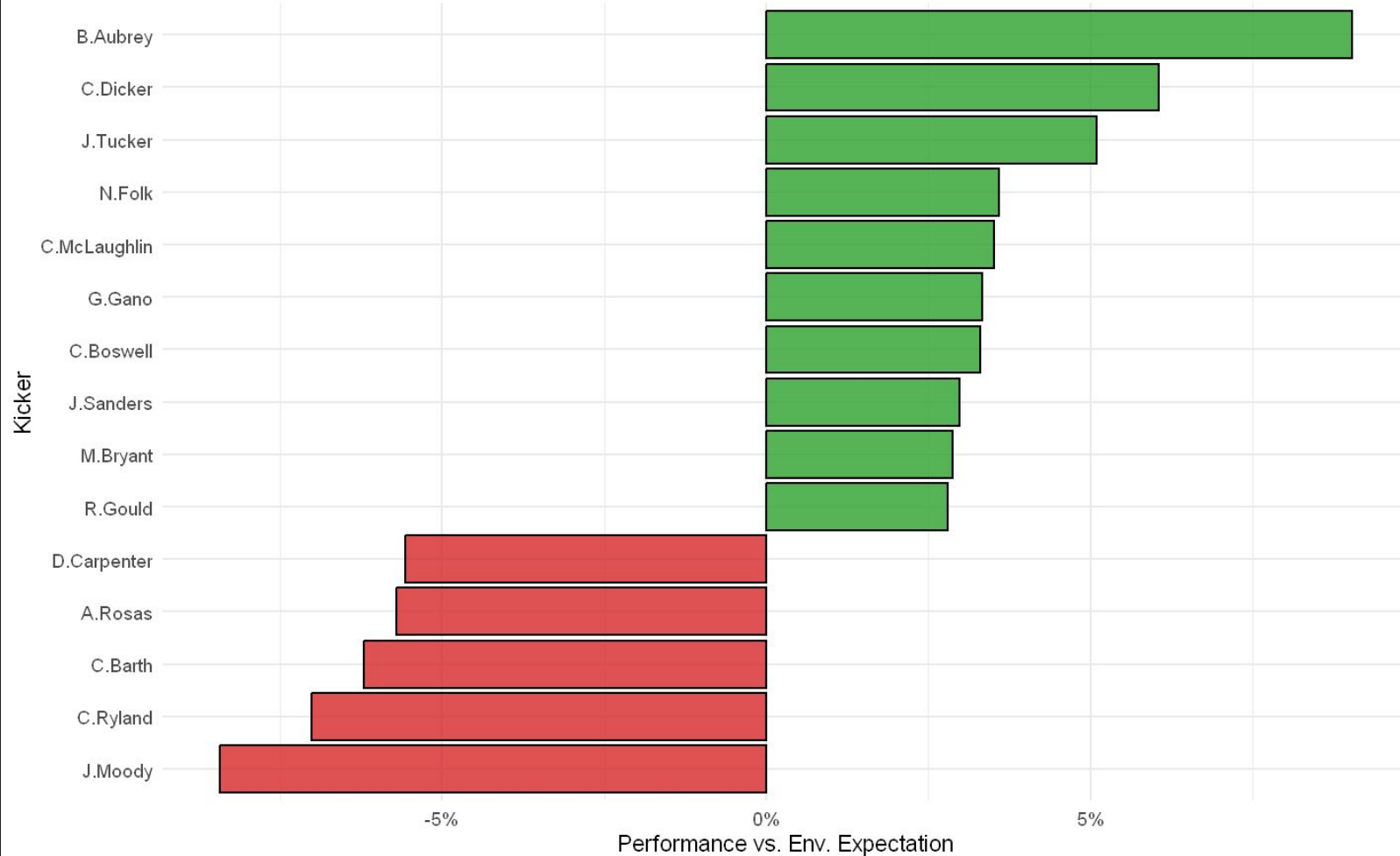
Takeaways

- **The Decision Zone:** The 45–55 yard range is "No Man's Land." This is where all play options converge
- Punting is inefficient (net ~20 yards), and Kicking is declining in value (3 points * declining make rate).
- **Rationality Confirmed:** Coaches decline kicks in this zone **not** because they lack confidence in the kicker's leg strength, but because **maximizing Win Probability requires aggression**.
- **Modeling Implication:** Treating these non-attempts as "Misses" (Pseudo-Labeling) is structurally incorrect. It penalizes the kicker for the coach's correct strategic choice to chase a Touchdown.



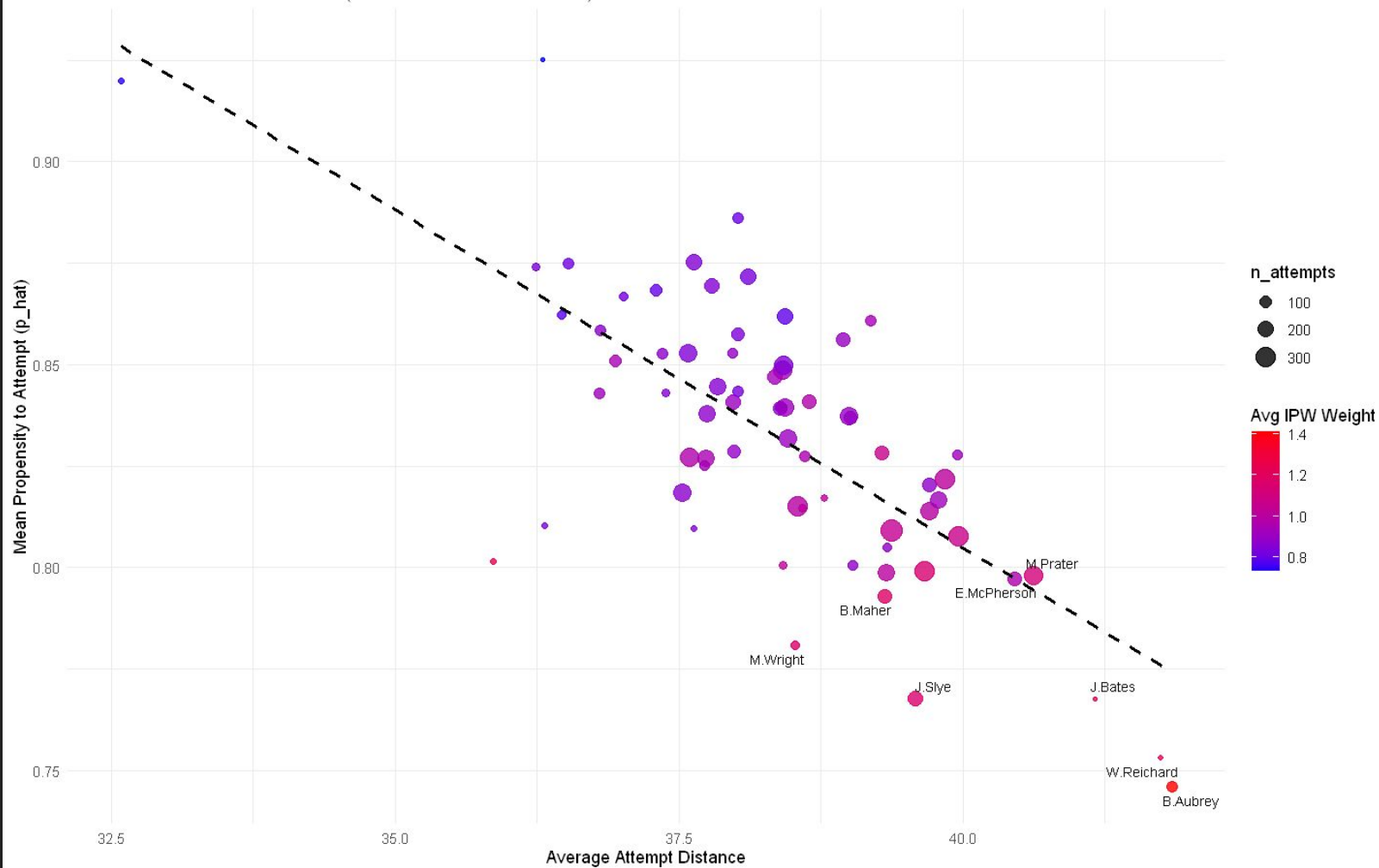
True Kicker Skill: FGOE (Environment Adjusted)

Baseline: M1 Agnostic (Distance + Weather + Stadium + Season, No Kicker ID)



Kicker Workload Analysis

Who takes the non-standard kicks? (Low Y-Axis = Harder Workload)



True Skill vs. Workload

Does high skill correlate with difficult assignments?

